AI GitHub Code Intelligence Agent

Project Description

The AI GitHub Code Intelligence Agent is an automated system that tracks repository commits and pull requests, evaluates them using AI models, detects risks, and provides intelligent analytics dashboards for engineering teams. It acts as an AI-powered code review assistant and engineering intelligence platform.

Core Capabilities

- Automated commit evaluation (quality, security, performance, complexity).
- Pull Request risk detection and architecture impact analysis.
- AI-generated code review summaries and suggestions.
- Automated PR commenting via GitHub App integration.
- Developer performance analytics and repository health scoring.

Phase 1 - Infrastructure Setup

- Setup NestJS with MongoDB (Mongoose).
- Configure Redis and BullMQ for background jobs.
- Create base Mongo schemas (Repository, Commit, PR, Evaluation).
- Setup React frontend with layout and routing.

Phase 2 - GitHub Integration

- Implement GitHub webhook endpoint.
- Validate webhook signature.
- Store commit and PR data in MongoDB.
- Push evaluation jobs to queue.

Phase 3 - AI Commit Evaluation

- Build AI evaluation service.
- Process commit jobs via queue.
- Store structured AI results in Evaluation collection.
- Display scores and risk levels in frontend.

Phase 4 - PR Intelligence

- Analyze diff size and breaking changes.
- Apply weighted risk scoring formula.
- Display architecture impact analysis in UI.

Phase 5 - GitHub App Automation

- Create GitHub App for secure integration.
- Auto-post AI review comments on PRs.
- Add configurable rule engine for risk thresholds.

Phase 6 - Analytics Dashboard

- Implement MongoDB aggregation for metrics.
- Developer leaderboard and risk trends.
- Repository health index and charts.

Phase 7 - Realtime & Scaling

- Add WebSocket for real-time evaluation updates.
- Add pagination, search, and filtering.
- Optimize MongoDB indexes.

Phase 8 - Production Hardening

- Rate limiting and security improvements.
- Logging and monitoring setup.
- Dockerization and deployment.